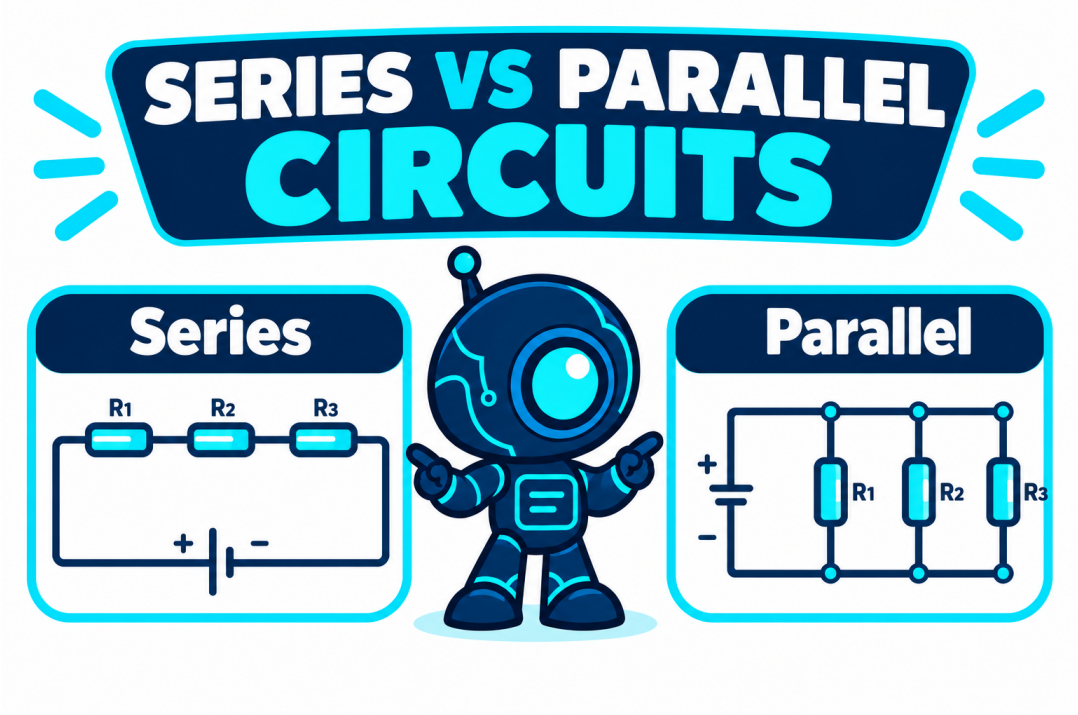
AI MECH LAB

Mechatronics Course — Beginner Track



Module 2, Topic 2.2

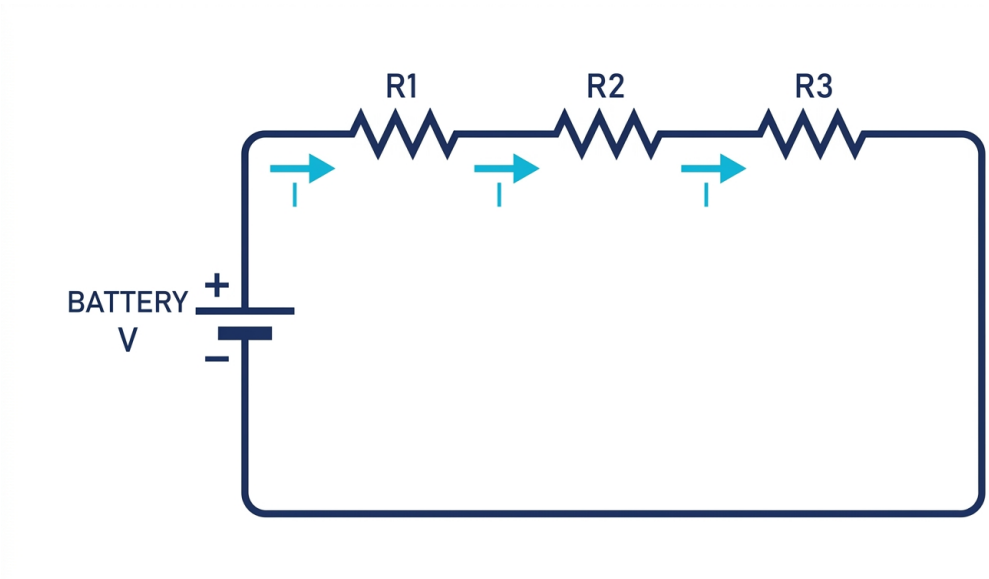
Series vs. Parallel Circuits

Lesson

You now understand voltage, current, resistance, and how Ohm's Law ties them together. But so far you've only looked at the simplest possible circuit: one source, one component, one loop. Real circuits almost always involve multiple components, and how you connect them — in series or in parallel — completely changes how voltage and current behave across the whole circuit. Getting this wrong is one of the most common reasons a beginner's circuit doesn't do what they expected.

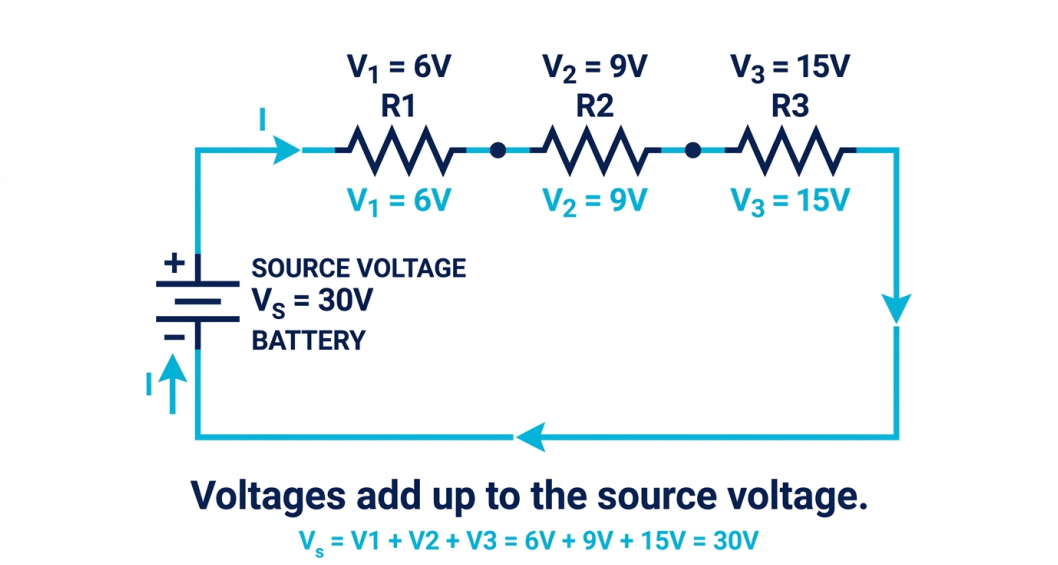
Series circuits: one path, shared current

In a series circuit, components are connected one after another, forming a single path for current to flow. There's only one route, so the same current flows through every component in that loop — current doesn't get "used up" or split along the way.



The same current flows through every component along a single series path.

Voltage behaves differently. The source voltage gets divided up across the components in the loop, in proportion to each component's resistance. If you add up the voltage across every component in a series circuit, it always equals the total source voltage — nothing is lost, it's just distributed.

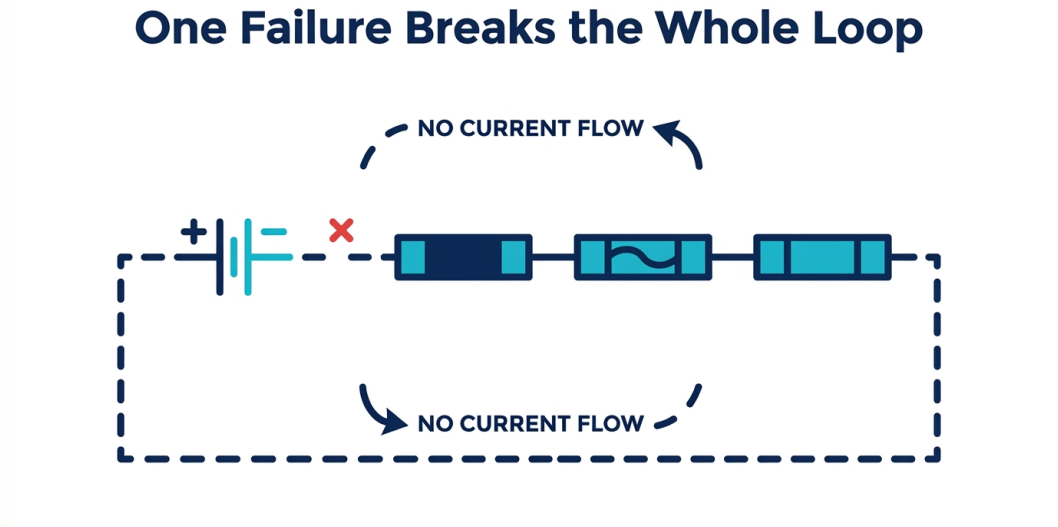


Voltage splits across series components but always sums back to the source voltage.

This is exactly what you saw in the LED and resistor example from Topic 2.1, without naming it yet: the 5V source split into 3V across the resistor and 2V across the LED, because they were connected in series, sharing one path and one current.

Series circuits: the single point of failure

Because there's only one path, series circuits have a real weakness: if any single component fails open (stops conducting), the entire loop breaks, and current stops everywhere in the circuit at once — not just at the failed component. This is the classic old-style string of holiday lights where one dead bulb takes out the whole strand.

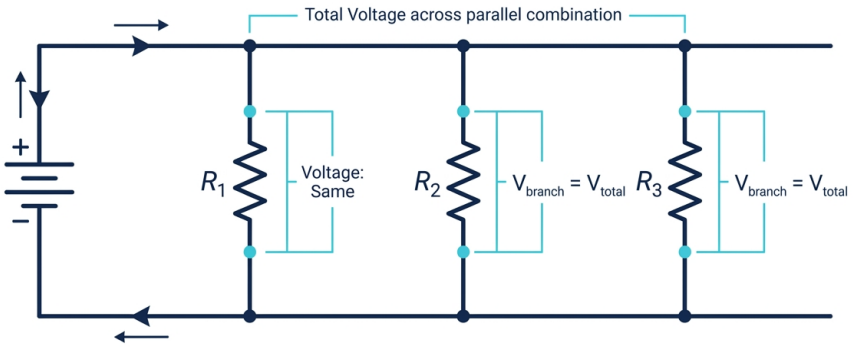


One failed component is enough to stop current everywhere in a series loop.

That weakness is sometimes intentional, though — a fuse is designed to be the deliberate weak point in a series path, breaking the circuit on purpose if current gets dangerously high.

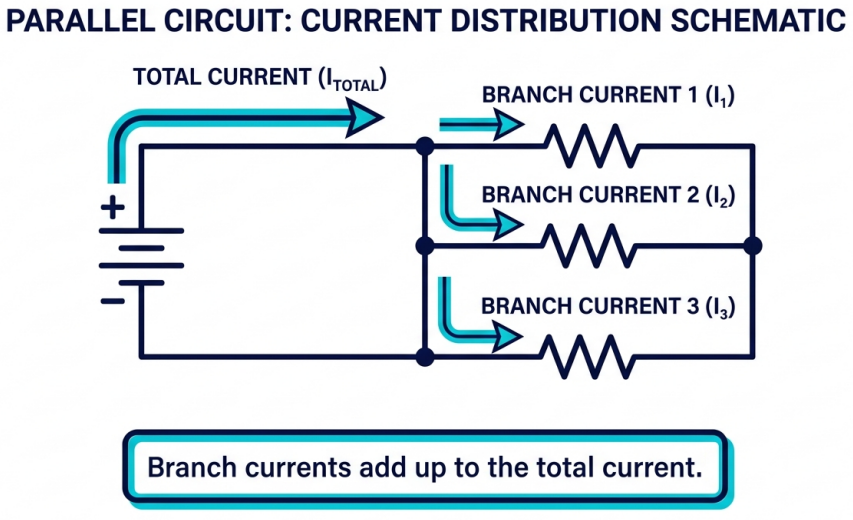
Parallel circuits: multiple paths, shared voltage

In a parallel circuit, components are connected across the same two points, each forming its own separate path back to the source. Voltage behaves simply here: every component in parallel sees the exact same voltage, because they're all connected across the same two points.



Every branch in a parallel circuit sees the same voltage.

Current behaves the opposite way from series: the total current from the source splits across the different paths, with each path carrying current in proportion to how much resistance it has. Add up the current through every parallel path, and it equals the total current leaving the source.

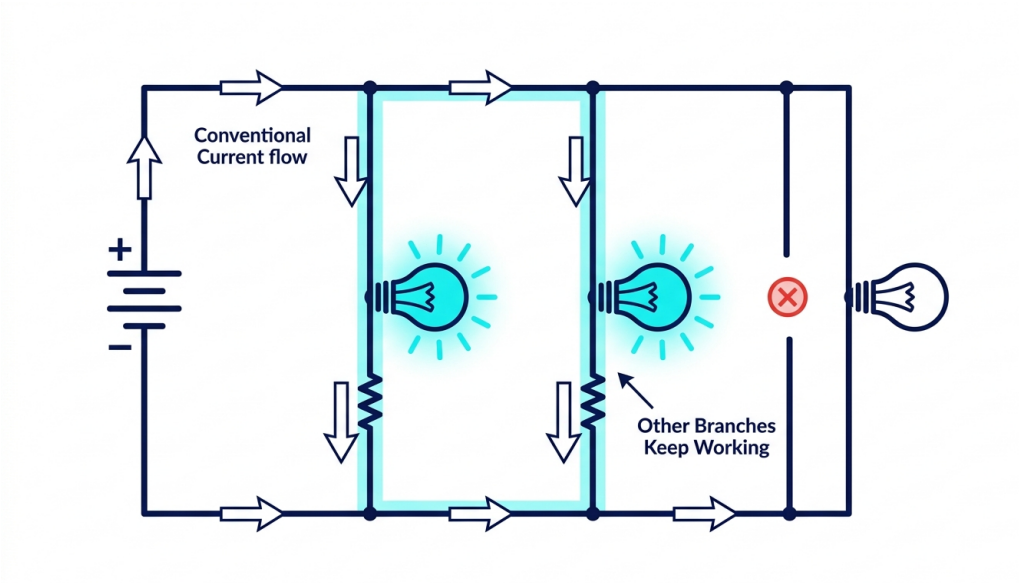


Branch currents in a parallel circuit add up to the total current from the source.

Parallel circuits: independence between branches

This is the practical advantage of parallel wiring: if one component in a parallel branch fails open, current simply stops flowing through that one branch — every other branch keeps working normally, because

each has its own separate path back to the source.



A failed branch in a parallel circuit doesn't affect the others.

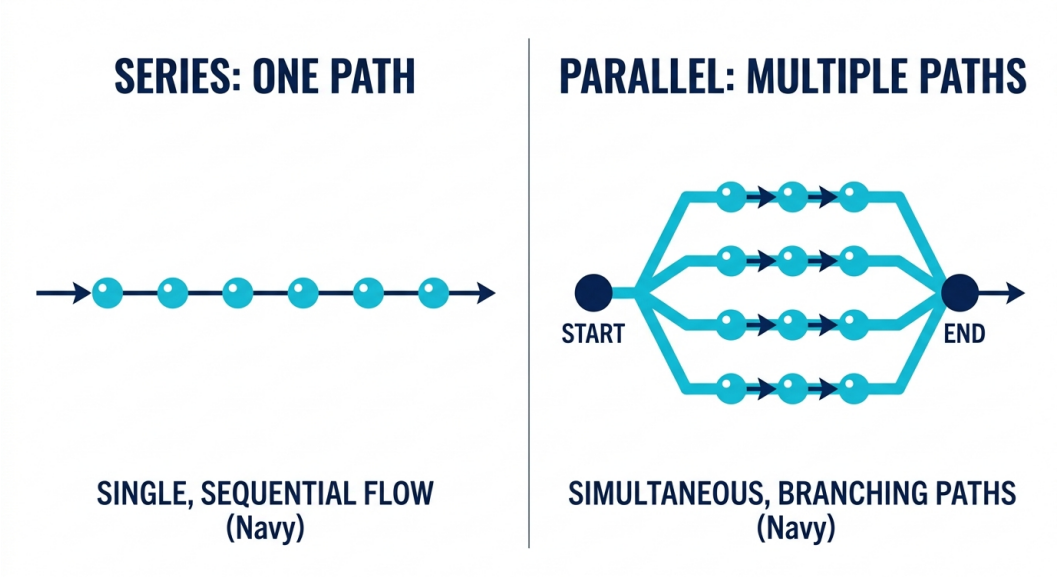
This is why household electrical wiring is parallel: one dead light bulb in your room doesn't take out every other outlet and light in the house.

Why this distinction matters for component design

This isn't just circuit theory — it directly affects decisions you'll make building real systems. Wiring multiple LEDs in series means they all share one current path, so if you're running them from a fixed-current source, each LED gets the same current, but the voltage needed from your source grows with every LED added. Wiring the same LEDs in parallel means each one sees full source voltage independently, but now your source needs to supply enough total current for all of them combined, and a single weak LED drawing more than its share can pull current away from the others.

Neither arrangement is universally "better" — which one you choose depends entirely on what your source can supply and what behavior you actually want from the circuit.

A quick way to tell them apart by eye

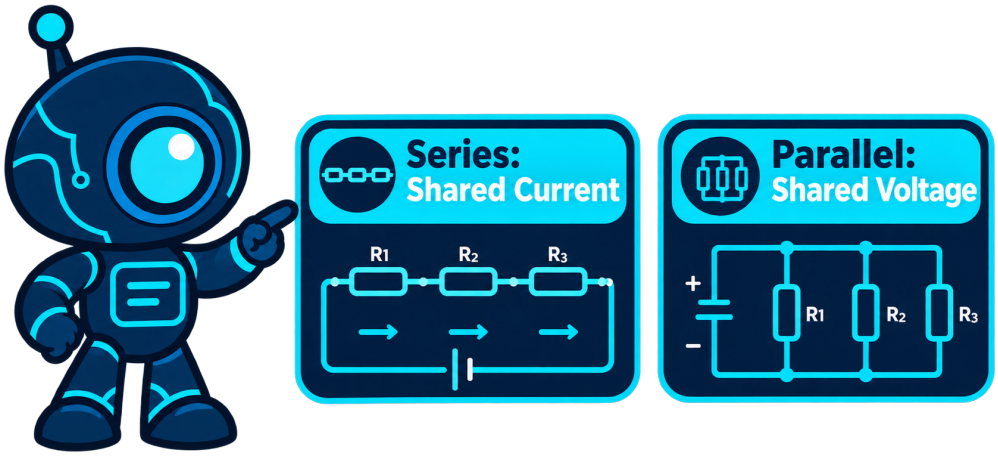


Series components form one line; parallel components branch and rejoin.

When you're looking at a circuit diagram, series components sit along one continuous line, like beads on a single string. Parallel components branch off the same two points and rejoin on the other side, like several lanes on a highway that all start and end at the same two junctions. Most real circuits you'll encounter later in this course combine both — some sections in series, some in parallel — so being able to recognize each pattern at a glance is a skill worth building now, before the circuits get more complex.

Where this leaves you

By the end of this lesson, you should be able to identify whether a given circuit arrangement is series or parallel, predict how voltage and current will behave in each case, and explain why a component failure affects a series circuit very differently than a parallel one. Next, you'll move from circuit arrangement to the individual building blocks themselves — resistors, capacitors, diodes, and transistors — the essential components you'll be wiring into both kinds of circuits throughout the rest of this course.



Series shares current; parallel shares voltage.